



Scenario Modeling and EMA for CAVs

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“All models are wrong,
but some are useful.”

– George Box, famous statistician



How can models be **USEFUL** in planning for CAVs?

- Scenario Planning
 - structured way for organizations to think about the future using a limited number of scenarios (e.g., best case, worst case, most likely, etc.)
- Exploratory Modeling Analysis (EMA)
 - simultaneously vary input assumptions across a wide range of future scenarios along key dimensions of uncertainty
 - to explore potential outcomes, find critical input assumptions, and identify future policy directions likely to be robust in the face of “deep uncertainty”



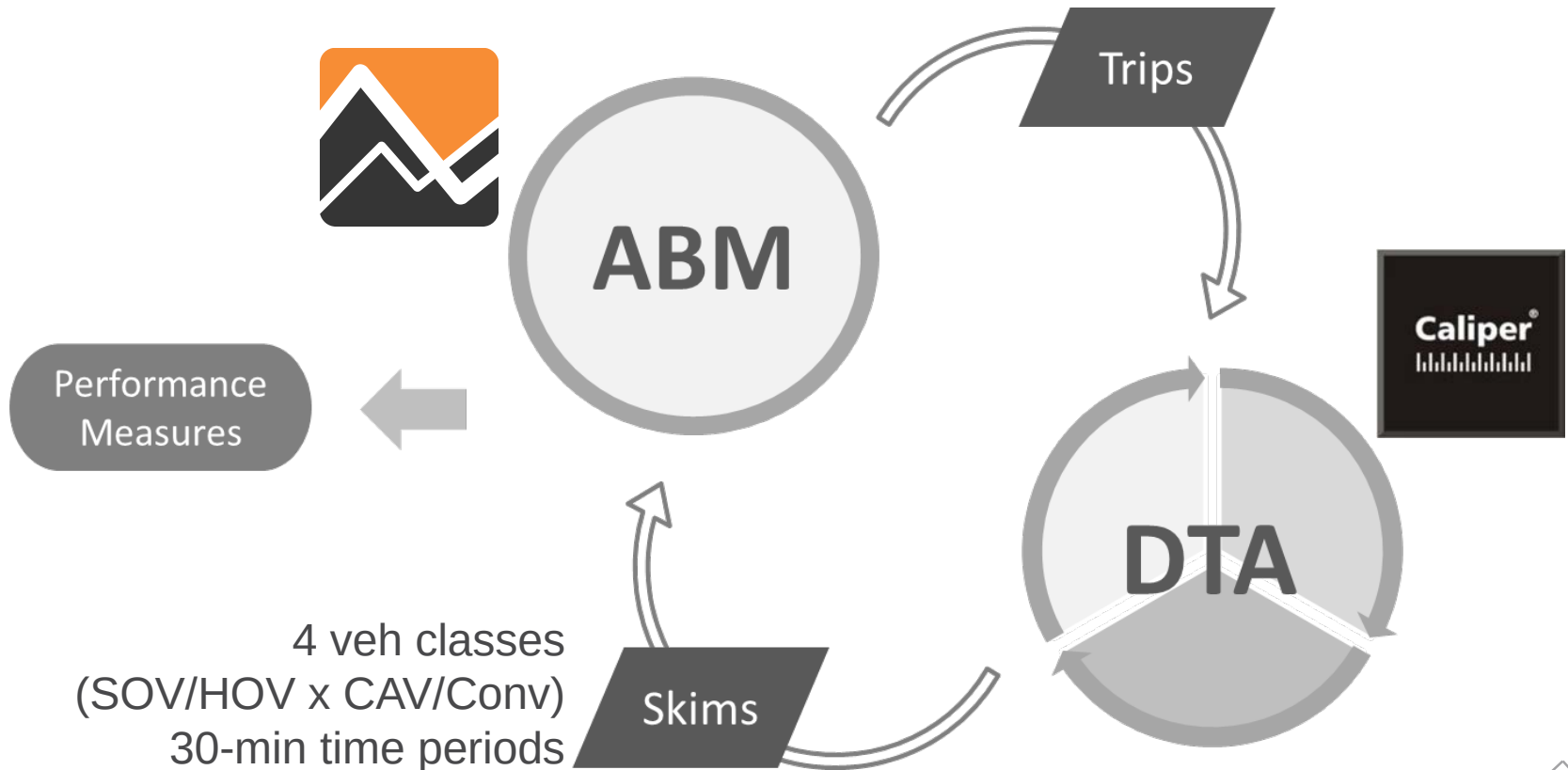


EMA with ABM-DTA

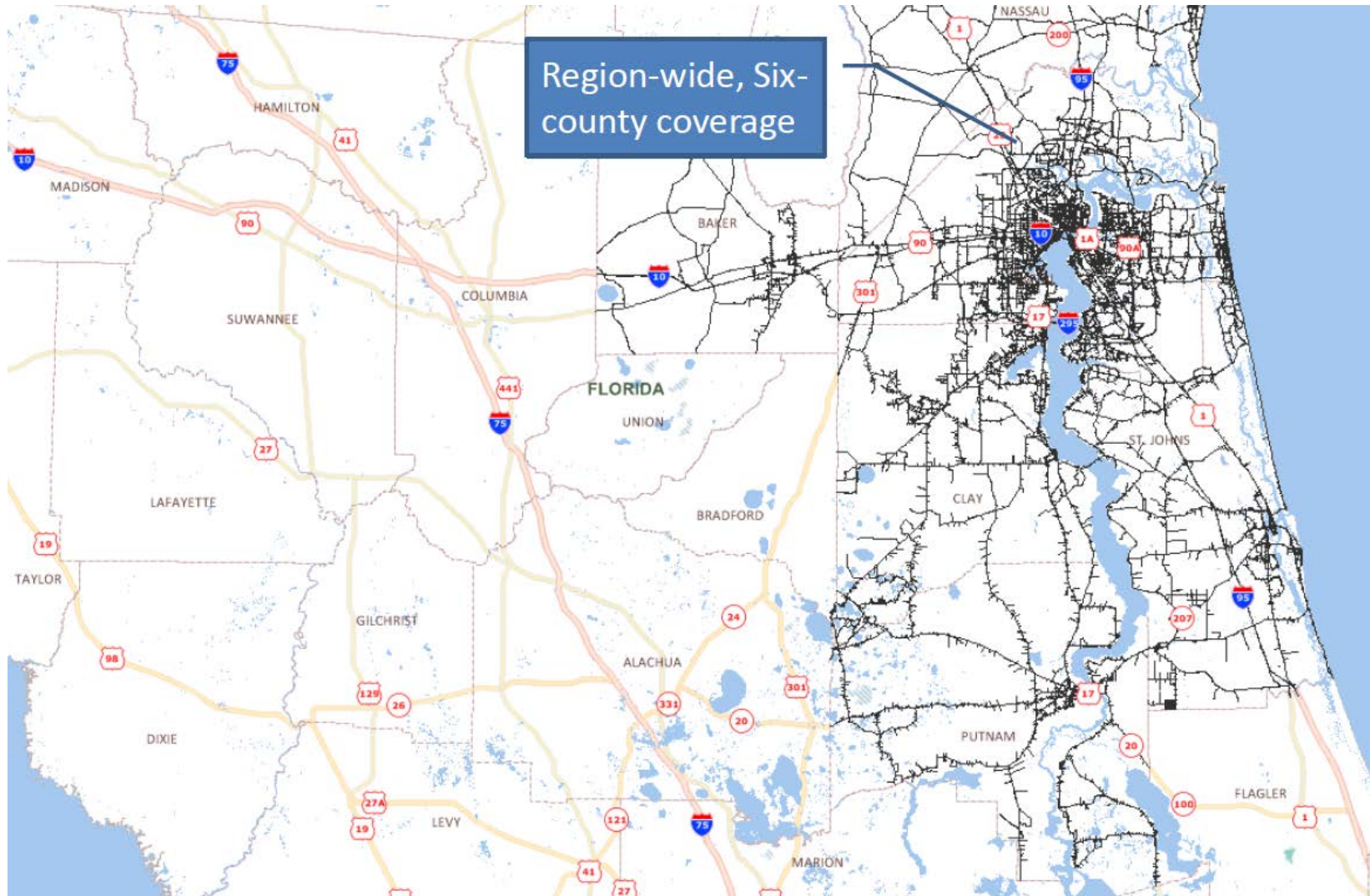
Jacksonville EMA Study



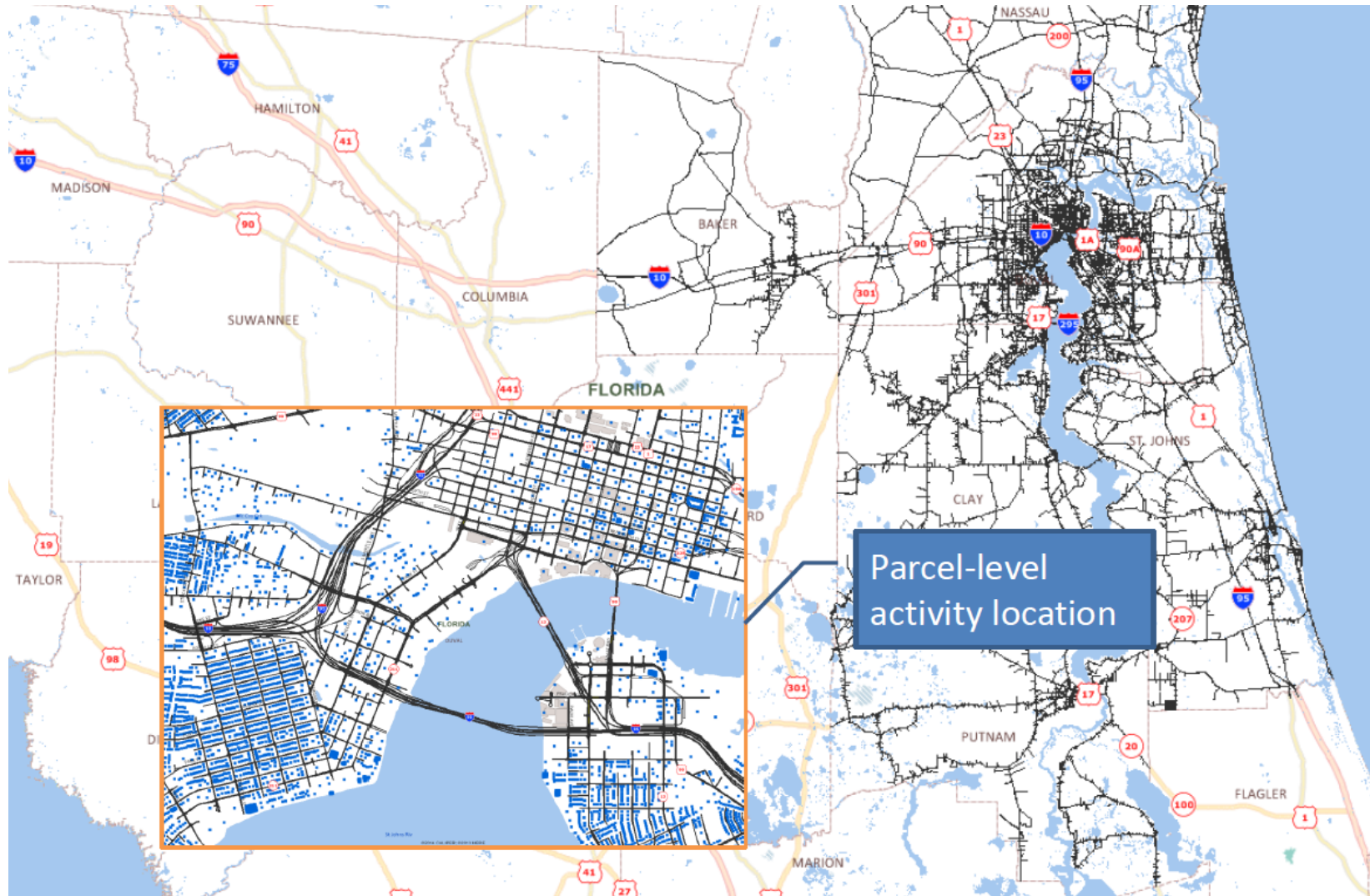
- Combined ABM-DTA model
- Similar, but shallower integration than 3C



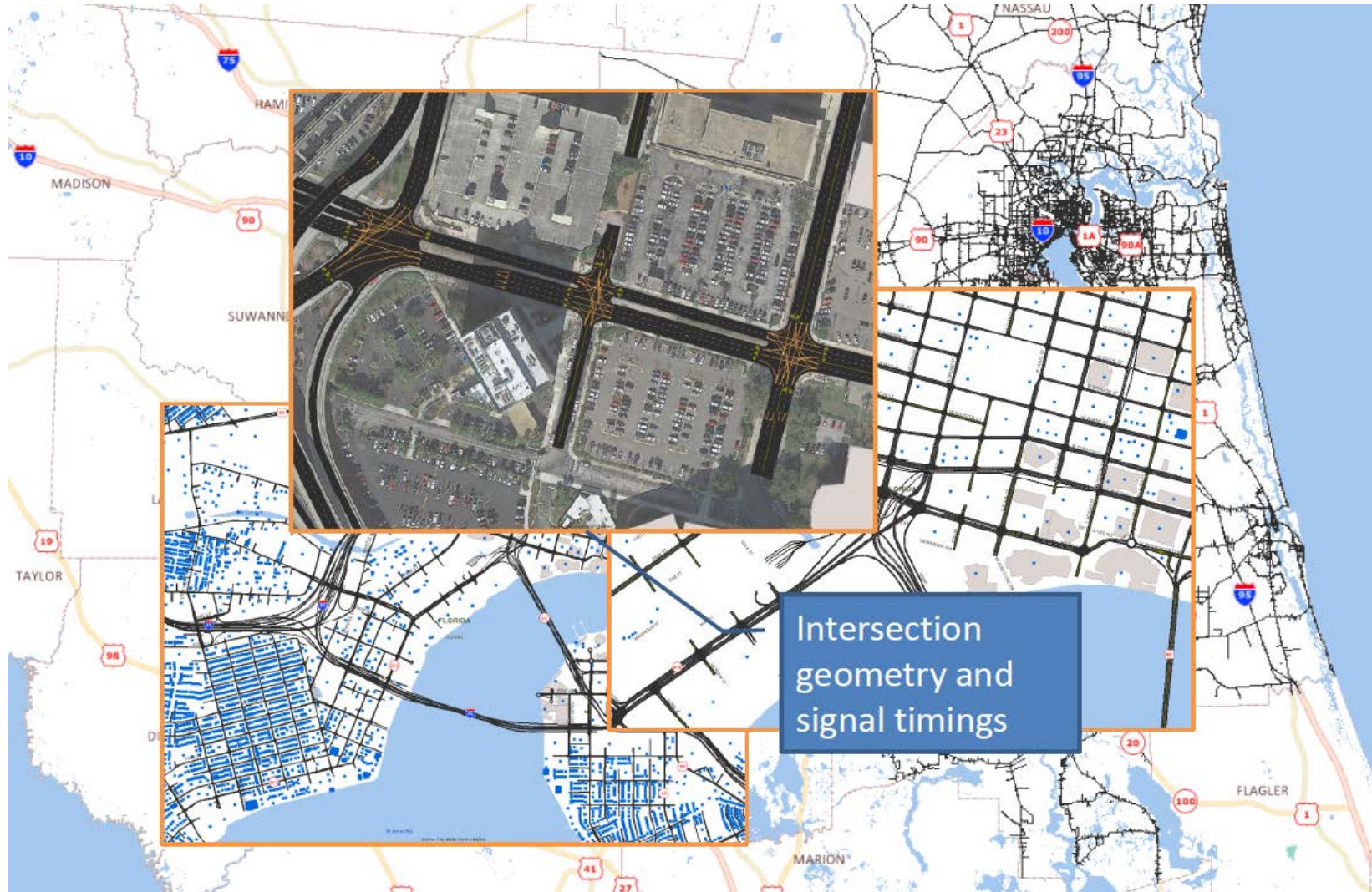
Jacksonville Model Overview



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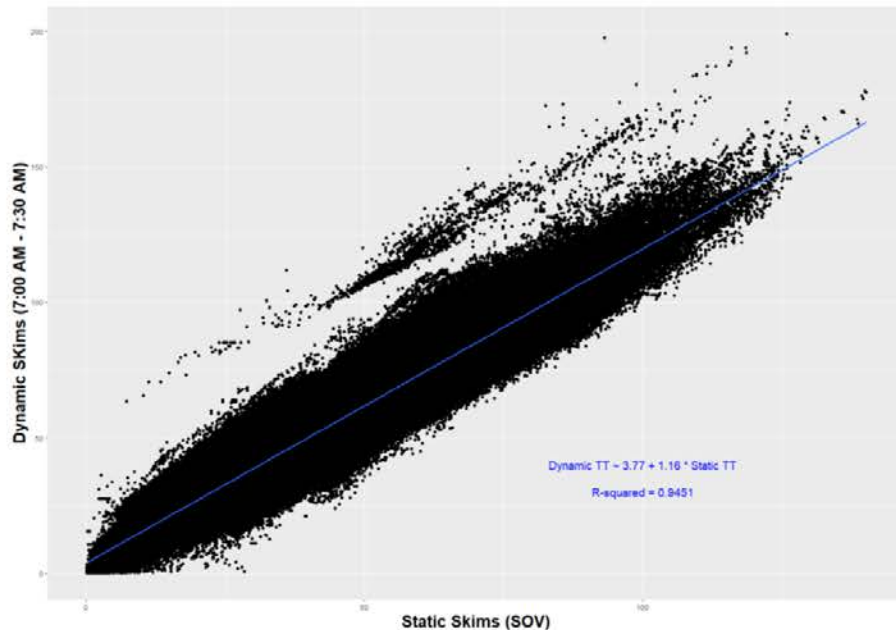


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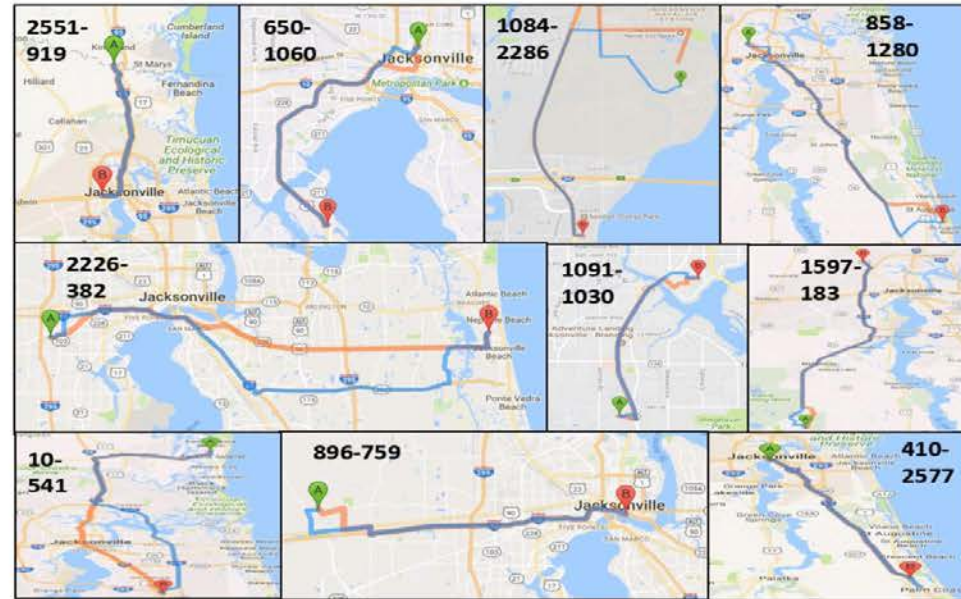


Validation of Dynamic Skims

Dynamic versus static



Outlier review



Model Runtime Performance

- Windows machines with 12 cores
 - TransModeler DTA: 5 - 9 AM, 25 iterations → 24 hours
 - DaySim ABM → 45 min
 - DaySim using AM dynamic skims + transpose for PM peak and static assignment for midday and night periods
 - Ran 3 to 5 feedback loops
 - Transit skims held constant
- Runtimes limited the number of EMA runs that could be done



CAV Enhancements to Daysim

- Auto ownership model includes choice between conventional and autonomous private vehicles
- The “paid rideshare” (TNC) mode added to mode choice
- TNCs can be specified to use AVs
- AV passengers can have lower disutility of travel time
- Use separate auto skim matrices for AVs
- ZOV Trips NOT included



CAV Enhancements to TransModeler

- Aspects of Driving Behavior Identified for Adaptation
 - Acceleration/deceleration
 - Car following headways
 - Choice of travel speed
 - Gap acceptance in lane changing
- Vehicle and Driving Behavior Assumptions
 - Removal of the random/human element from aspects controlled by the vehicle
 - Aspects deterministic, predictable, homogeneous



Experimental Design

SCENARIO	PRIVATE AV ADOPTION	SHARED AV ADOPTION	RESERVED AV CAPACITY	AUTOMATION LEVEL
BB-N0	None	None	None	None
MM-L3	Medium	Medium	Interstate left lanes	Level 3
MM-AC	Medium	Medium	None	Level 3 + ACC
MM-LC	Medium	Medium	Interstate left lanes	Level 3 + ACC
MM-IC	Medium	Medium	Interstate all lanes (only inside the I 295 ring road)	Level 3 + ACC
LH-L3	Low	High	Interstate left lanes	Level 3
LH-AC	Low	High	None	Level 3 + ACC
LH-LC	Low	High	Interstate left lanes	Level 3 + ACC
LH-IC	Low	High	Interstate all lanes (only inside the I 295 ring road)	Level 3 + ACC
HL-L3	High	Low	Interstate left lanes	Level 3
HL-AC	High	Low	None	Level 3 + ACC
HL-LC	High	Low	Interstate left lanes	Level 3 + ACC
HL-IC	High	Low	Interstate all lanes (only inside the I 295 ring road)	Level 3 + ACC
HH-L3	High	High	Interstate left lanes	Level 3
HH-AC	High	High	None	Level 3 + ACC
HH-LC	High	High	Interstate left lanes	Level 3 + ACC
HH-IC	High	High	Interstate all lanes (only inside the I 295 ring road)	Level 3 + ACC



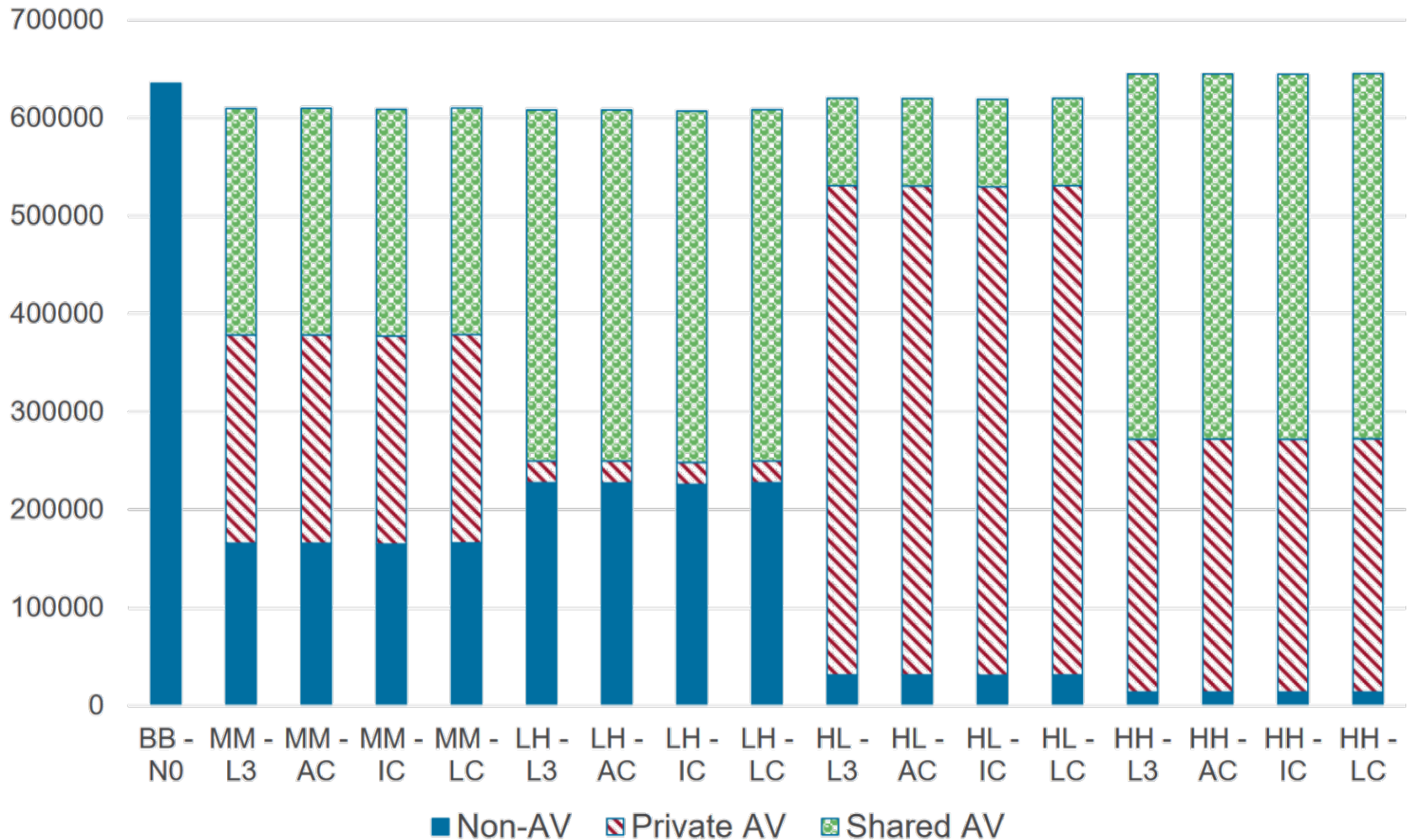
Ran 3 Global Iterations to Reasonable Convergence

Change in overall predicted average trip speeds from iteration 2 to iteration 3

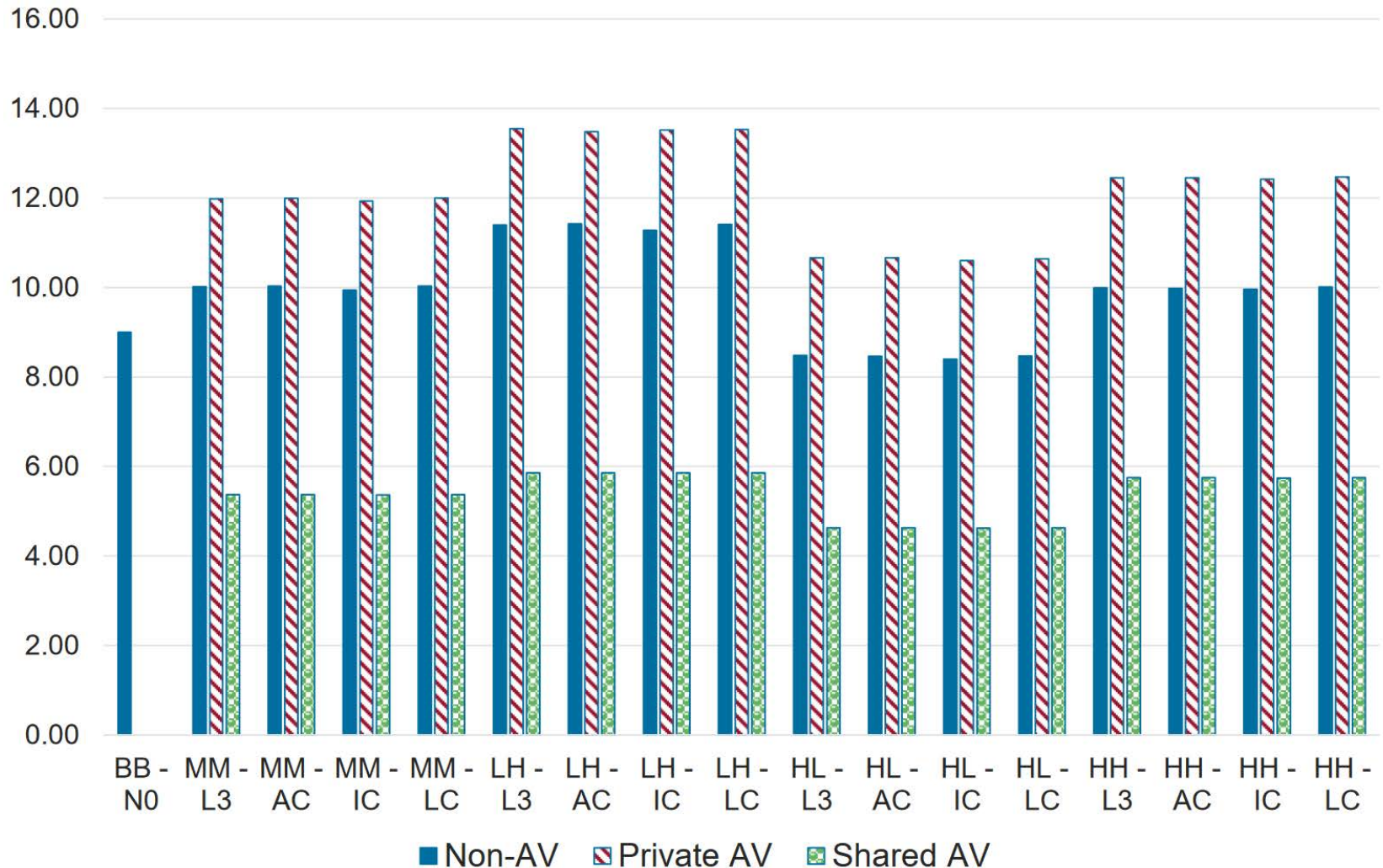
Run	5:00 am– 5:29 am	5:30 am– 5:59 am	6:00 am– 6:29 am	6:30 am– 6:59 am	7:00 am– 7:29 am	7:30 am– 7:59 am	8:00 am– 8:29 am	8:30 am– 8:59 am
BB–N0	0.13%	-0.13%	0.09%	0.23%	0.16%	0.00%	0.24%	0.29%
MM–L3	-0.07%	0.17%	-0.31%	-0.16%	-0.25%	-0.11%	-0.70%	-1.17%
MM–AC	0.04%	-0.04%	0.27%	0.44%	0.39%	0.15%	-0.07%	-0.13%
MM–IC	0.26%	0.04%	-0.26%	0.02%	0.34%	-0.07%	-0.32%	-0.45%
MM–LC	0.15%	-0.11%	0.33%	0.33%	0.45%	0.49%	0.47%	0.67%
LH–L3	-0.11%	-0.11%	0.12%	0.16%	0.06%	0.73%	0.34%	0.13%
LH–AC	-0.22%	0.04%	-0.19%	-0.04%	-0.18%	-0.09%	-0.13%	0.22%
LH–IC	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
LH–LC	-0.17%	0.07%	0.27%	0.14%	0.10%	0.64%	0.70%	0.58%
HL–L3	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
HL–AC	-0.17%	0.06%	0.35%	0.16%	0.46%	0.22%	0.37%	-0.09%
HL–IC	0.17%	0.04%	-0.28%	-0.08%	0.13%	0.18%	-0.23%	-0.46%
HL–LC	-0.22%	-0.11%	-0.17%	-0.31%	-0.04%	-0.51%	-0.69%	-1.34%
HH–L3	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
HH–AC	-0.28%	0.00%	0.14%	-0.14%	0.19%	0.18%	0.59%	0.21%
HH–IC	0.15%	0.00%	-0.12%	-0.08%	0.04%	0.04%	0.09%	-0.26%
HH–LC	0.00%	-0.04%	-0.12%	0.12%	0.38%	0.28%	0.51%	0.44%



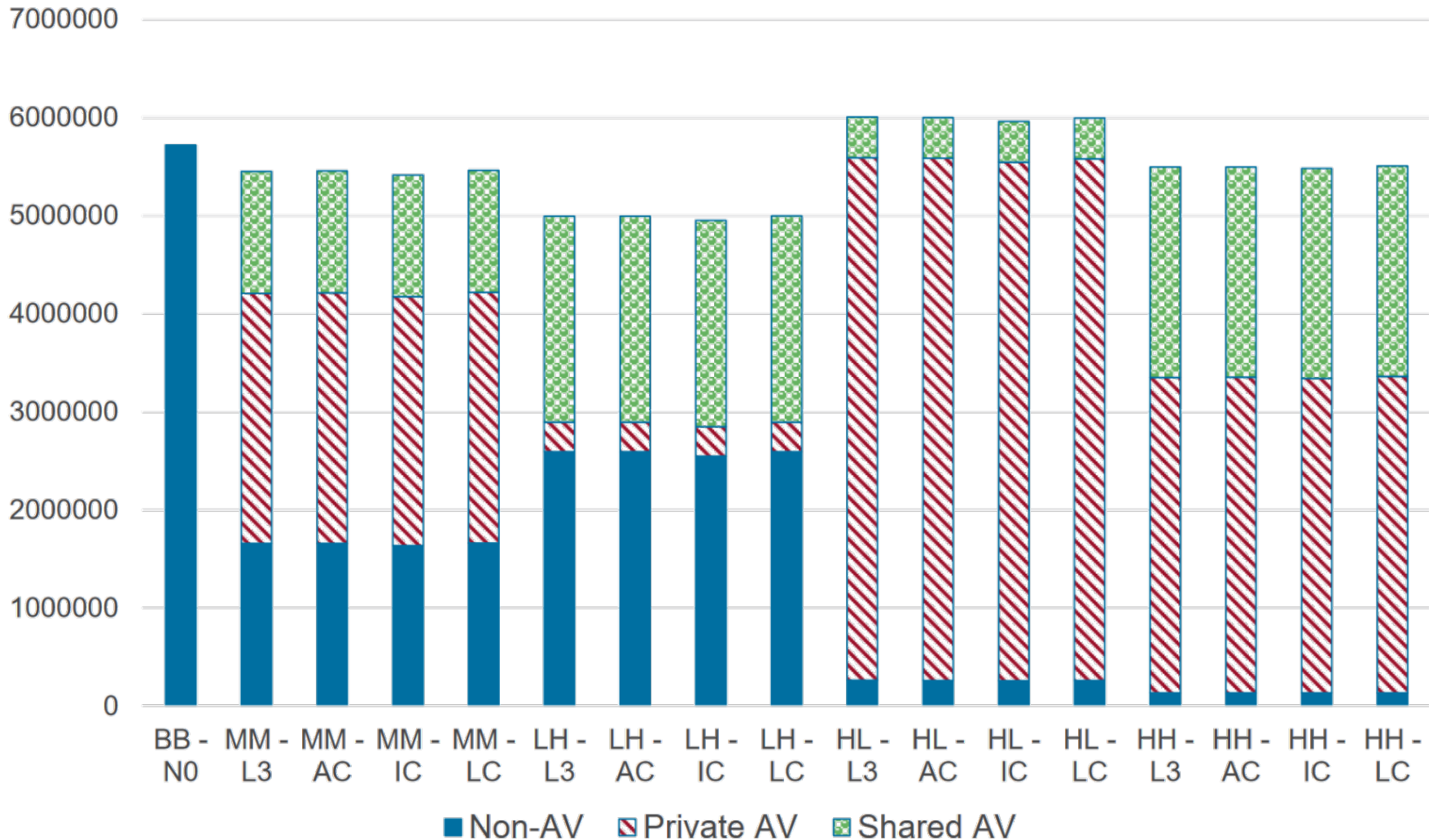
AM Vehicle-Trips, by Vehicle Type and Scenario



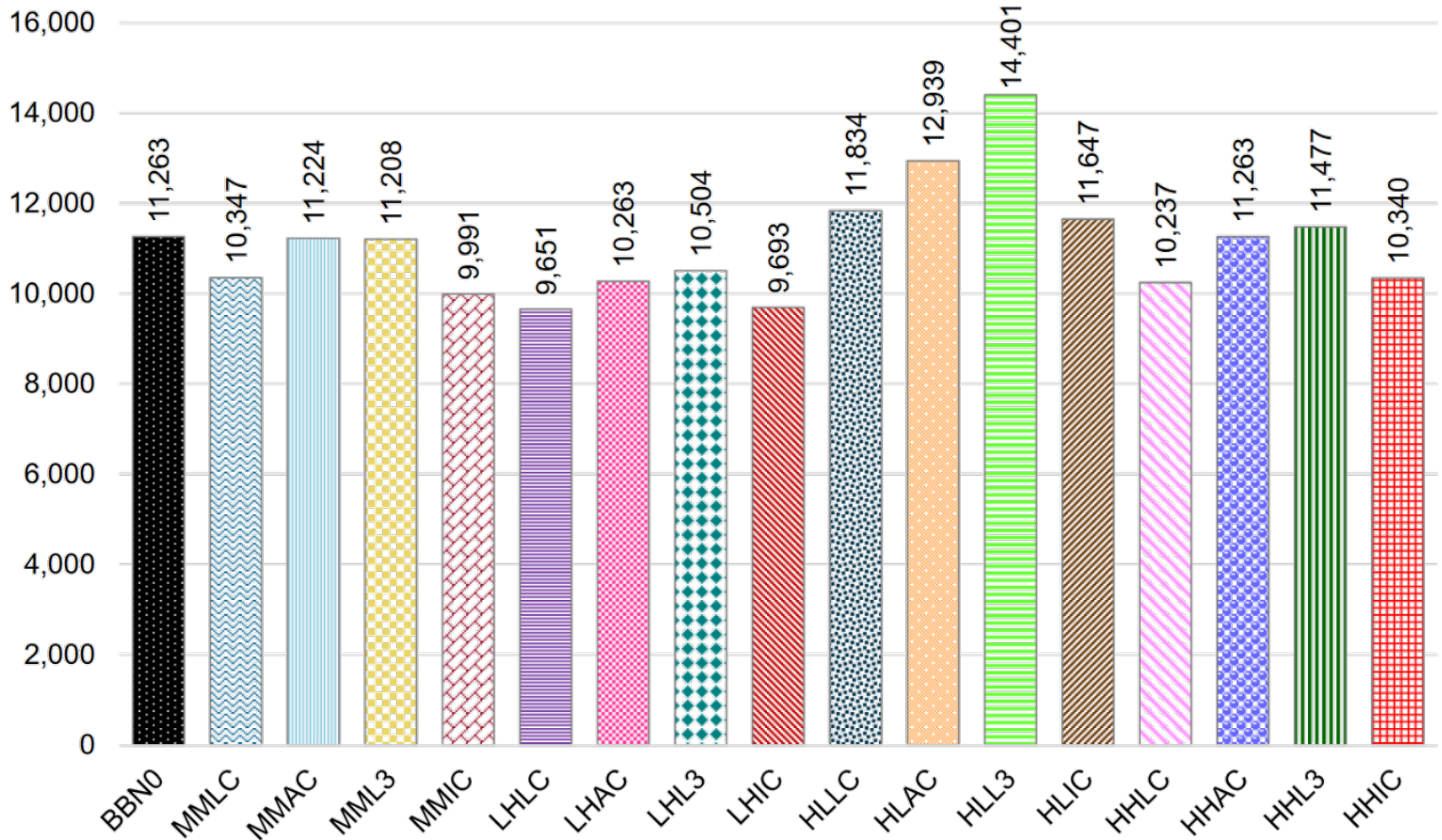
AM Average Vehicle-Trip Distances



AM VMT, by Vehicle Type and Scenario



DTA Vehicle-Hours of Delay, by Scenario



Regression Model on ABM Output: Total VMT (millions), by Scenario / Time Period / Vehicle Type

Vehicle Type	Non-AV	Non-AV	Private AV	Private AV	Shared AV	Shared AV	All types	All types
Variables	Coeff.	T-stat	Coeff.	T-stat	Coeff.	T-stat	Coeff.	T-stat
Constant	0.262	11.1	0.443	10.6	0.226	12.9	0.931	117.6
Demand - High Private, Low Shared	-0.174	-9.8	0.346	11.0	-0.103	-7.8	0.068	11.4
Demand - Low Private, High Shared	0.116	6.5	-0.281	-8.9	0.108	8.1	-0.057	-9.6
Demand - High Private, High Shared	-0.190	-10.6	0.083	2.6	0.113	8.5	0.006	1.1
Supply - Network scenario AC	0.000	0.0	0.000	0.0	0.000	0.0	0.000	0.0
Supply - Network scenario IC	-0.002	-0.1	-0.002	-0.1	0.000	0.0	-0.004	-0.7
Supply - Network scenario LC	0.000	0.0	0.000	0.0	0.000	0.0	0.000	0.1
Arrive Period - 5:00 to 5:29	-0.182	-7.2	-0.434	-9.7	-0.237	-12.7	-0.853	-100.7
Arrive Period - 5:30 to 5:59	-0.177	-7.0	-0.422	-9.5	-0.231	-12.3	-0.830	-98.1
Arrive Period - 6:00 to 6:29	-0.051	-2.0	-0.109	-2.5	-0.075	-4.0	-0.235	-27.8
Arrive Period - 6:30 to 6:59	-0.057	-2.3	-0.125	-2.8	-0.081	-4.3	-0.263	-31.1
Arrive Period - 7:00 to 7:29	0.035	1.4	0.107	2.4	0.051	2.7	0.192	22.7
Arrive Period - 7:30 to 7:59	0.008	0.3	0.042	0.9	0.026	1.4	0.076	9.0
Arrive Period - 8:30 to 8:59	-0.017	-0.7	-0.048	-1.1	-0.018	-1.0	-0.083	-9.8



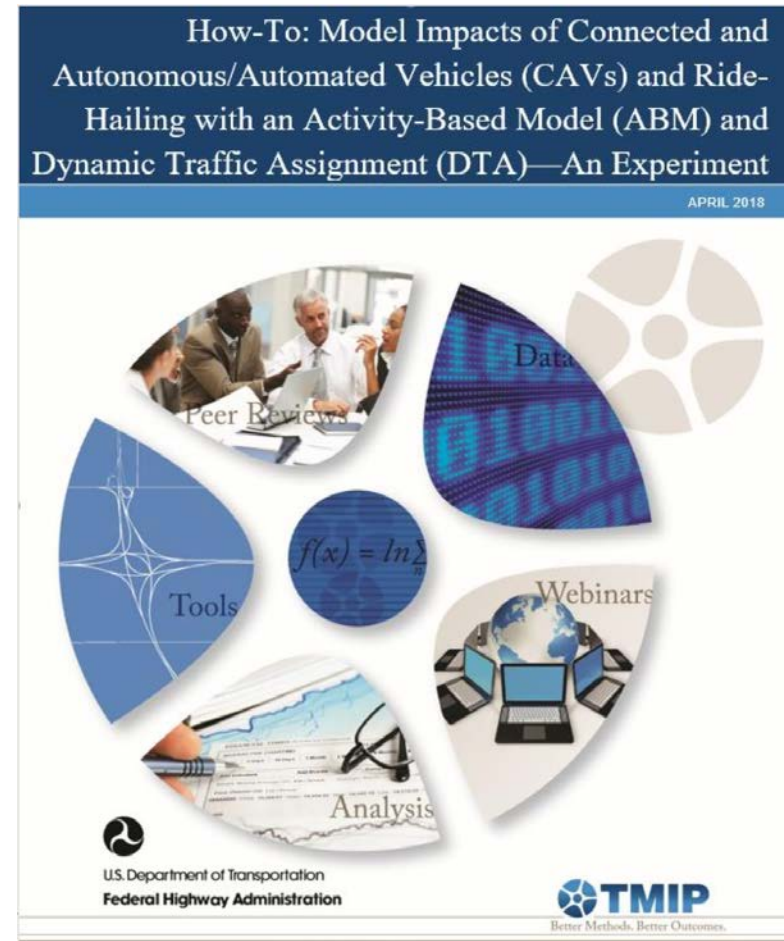
Regression Model on DTA Output: Average Trip Speed (MPH), by Scenario / Time Period / Vehicle Type

Vehicle Type	Non-AV	Non-AV	AV	AV	Both types	Both types
Variables	Coeff.	T-stat	Coeff.	T-stat	Coeff.	T-stat
Constant	31.292	111.2	31.070	136.0	31.036	136.9
Demand - High Private, Low Shared	-1.138	-5.4	-0.608	-3.5	-0.574	-3.3
Demand - Low Private, High Shared	0.618	2.9	-0.533	-3.1	-0.007	0.0
Demand - High Private, High Shared	0.455	2.1	0.135	0.8	0.206	1.2
Supply - Network scenario AC	1.064	5.0	0.004	0.0	0.328	1.9
Supply - Network scenario IC	-0.024	-0.1	1.416	8.2	1.008	5.9
Supply - Network scenario LC	0.724	3.4	0.975	5.6	0.943	5.5
Arrive Period - 5:00 to 5:29	11.496	38.2	11.898	48.7	11.829	48.8
Arrive Period - 5:30 to 5:59	13.737	45.7	14.314	58.6	14.258	58.8
Arrive Period - 6:00 to 6:29	11.052	36.7	11.193	45.8	11.306	46.7
Arrive Period - 6:30 to 6:59	8.516	28.3	8.963	36.7	8.949	36.9
Arrive Period - 7:00 to 7:29	4.976	16.5	4.779	19.6	4.888	20.2
Arrive Period - 7:30 to 7:59	1.651	5.5	1.753	7.2	1.783	7.4
Arrive Period - 8:30 to 8:59	-0.573	-1.9	0.156	0.6	0.011	0.0



EMA Summary

- Assumptions
 - Level 3
 - No ZOVs
 - AM peak only
 - Uncertainty in levels of private and shared CAVs, dedicated lanes
- Results
 - Delay varied from -14% to +28%
 - Private CAVs increased VMT
 - Shared CAVs decreased VMT
 - Dedicated left lane almost as good as dedicating all freeway lanes



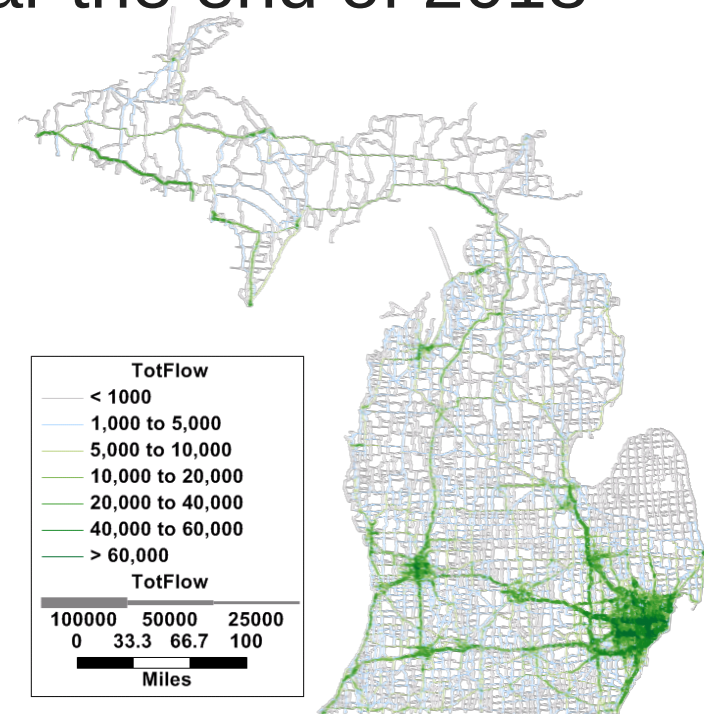


A Trip-Based Framework for CAVs

The New Michigan Statewide Model



- One of the longest traditions of statewide modeling in the country since early 1970's
- Last major update was mid 1990's
- New model complete near the end of 2018
 - Data-driven approach using AirSage & ATRI
 - Advanced trip-based passenger modeling
 - Linkage of HB & NHB, LD and Visitor trips
 - Commodity-flow based freight modeling
 - Summer model



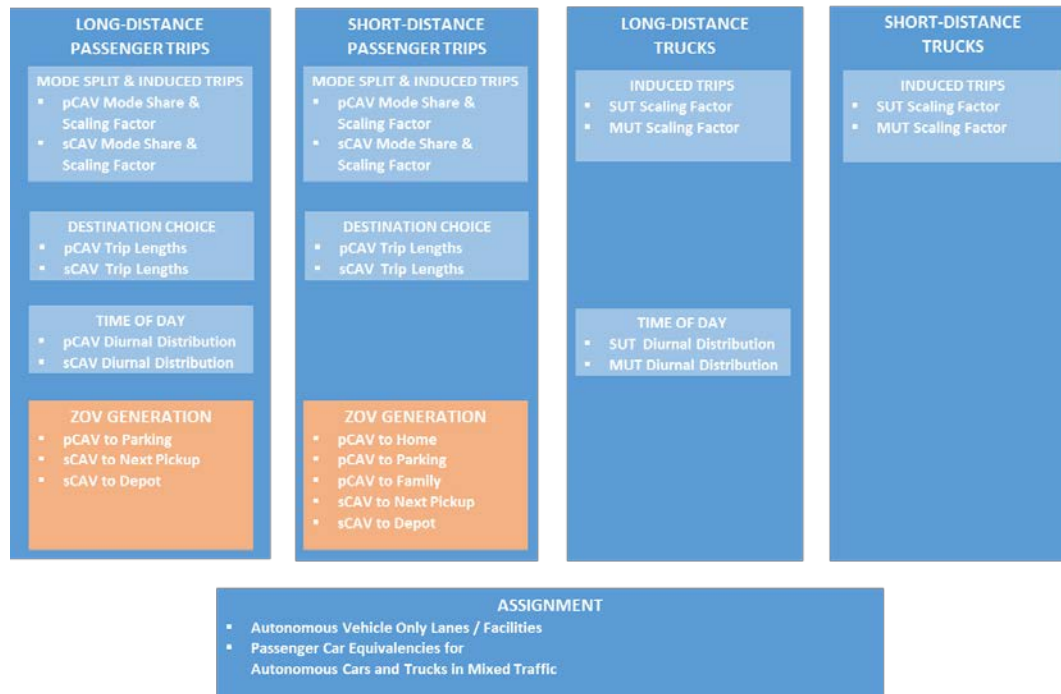
CAVs in Michigan

- Home of the Auto Industry
 - All auto-makers actively investing in CAV technology
- Public-Private Partnerships
 - Research rapidly moving into reality
 - U of M has been using autonomous shuttles for almost a year



CAVs in Trip-Based Models

- Lots of off-model / manual analysis
- Michigan, Illinois statewide models
- New factors / market segments + ZOV step



CAVs in Trip-Based MPO Models

- Lots of off-model / manual analysis
- Adapted Michigan framework
- Charleston, Charlottesville, & Ann Arbor MPOs

RESIDENT INTERNAL PASSENGER TRIPS

AUTO OWNERSHIP

- Overall Ownership Level
- Split of Vehicles between CAV vs. Conventional

INDUCED TRIPS

- Adjust Trip Rates by Seniors, Children, Autos

DESTINATION CHOICE

- Adjust Trip Lengths by CAV ownership

MODE CHOICE

- Add MaaS modes
- Add CAV / Conv. Submodes

ZOV GENERATION

- HH CAV to Family
- HH CAV to Home
- HH CAV to Free Parking
- HH CAV to Circulate
- MaaS CAV to Next Pickup
- MaaS CAV to Depot

EXTERNAL AND TRUCK TRIPS

INDUCED TRIPS

- EE & EI/IE Scaling Factors
- SUT Scaling Factor
- MUT Scaling Factor

TIME OF DAY

- Ext Diurnal Distributions
- Trk Diurnal Distributions

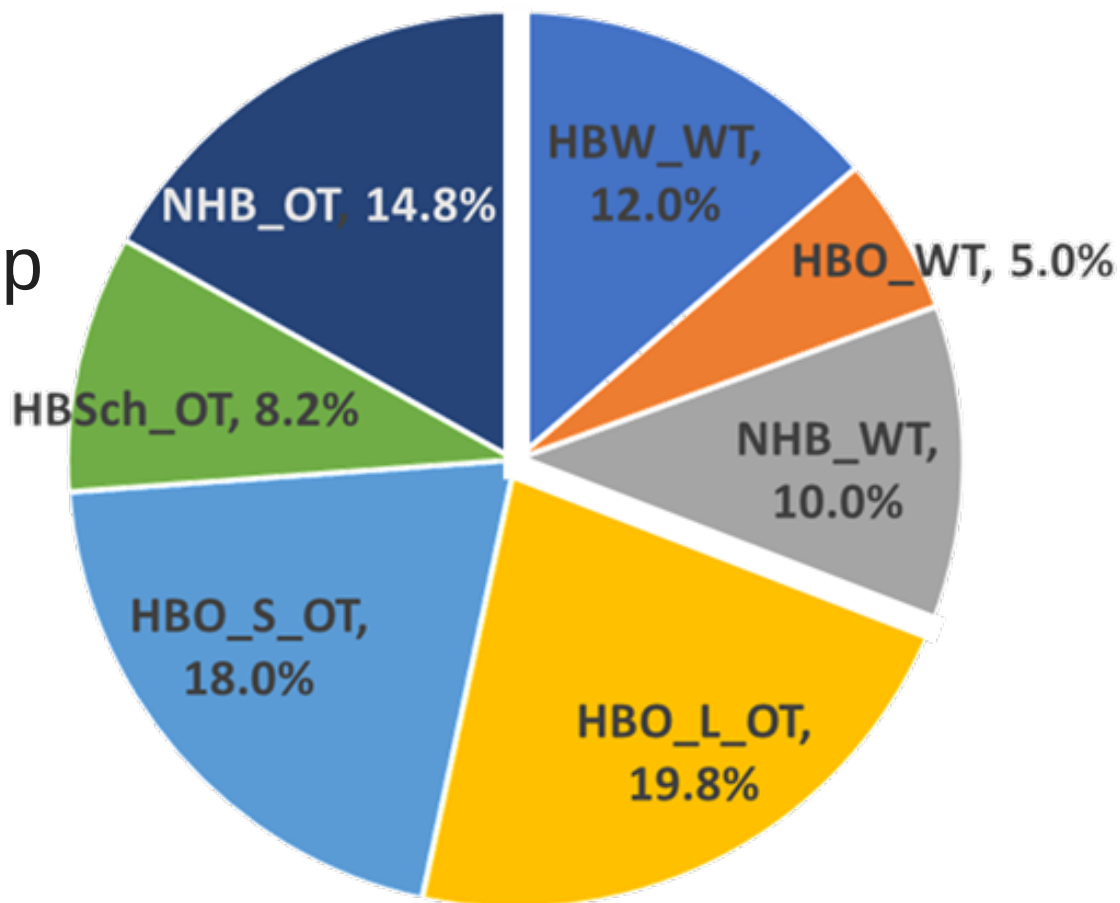
ASSIGNMENT

- Autonomous Vehicle Only Lanes / Facilities
- Passenger Car Equivalencies for Autonomous Cars and Trucks in Mixed Traffic



Model Segmentation

- Vehicle Ownership
 - No vehicles
 - Vehicles < Adults
 - Without CAV
 - With CAV
 - Vehicles $\geq$ Adults
 - Without CAV
 - With CAV
- Non-Work Activities segmented by DURATION
 - Less than 30 min (won't send vehicle home)
 - Greater than 30 min (might send vehicle home)



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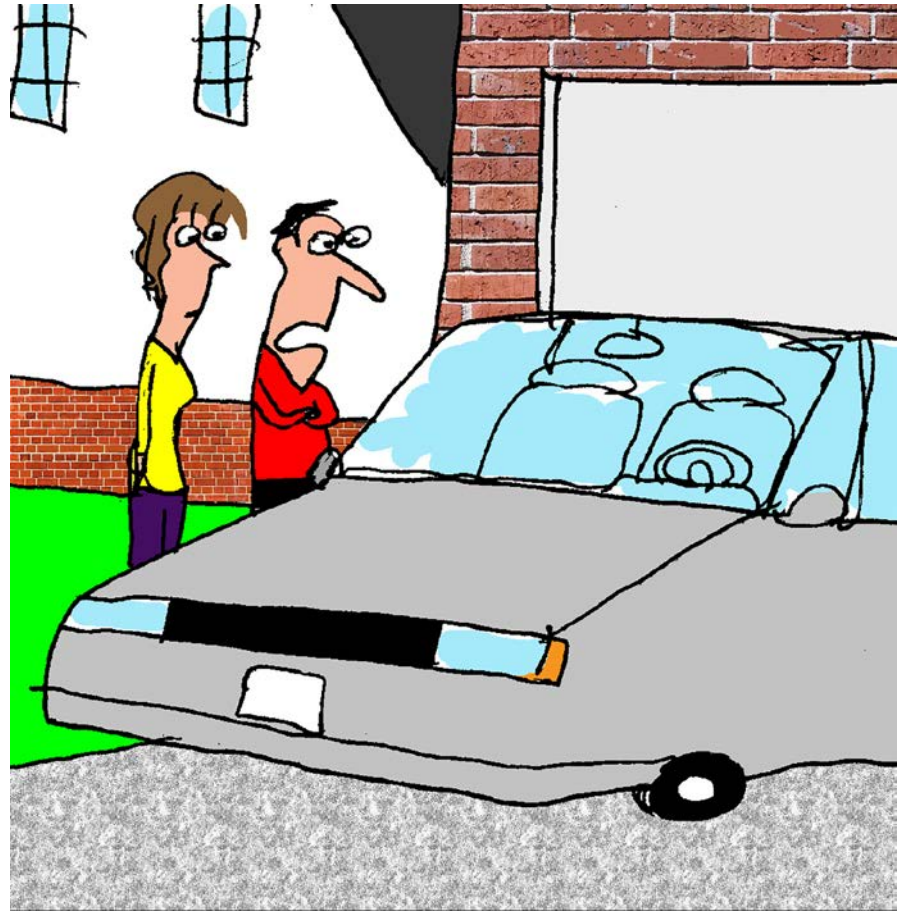
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Auto Ownership

- Subdivide HH autos into conventional and CAV by income
- Decrease overall ownership



"Maybe I can buy a self driving car, and hire it out to Uber to make the payments."

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Trip Generation

- Scale up trips to represent induced demand
- Largest increases to households with:
 - Disabled
 - Seniors
 - Children
- More long distance / external trips from reduced lodging cost?
- [Also, careful of trip rates by vehicle ownership]



Source: Jalopnik.com

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Trip Distribution



- Passengers may be willing to travel farther since time in CAVs can be used positively for working, relaxing, sleeping, etc.
- User can factor down traveler sensitivity to travel time / impedance

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Mode Choice

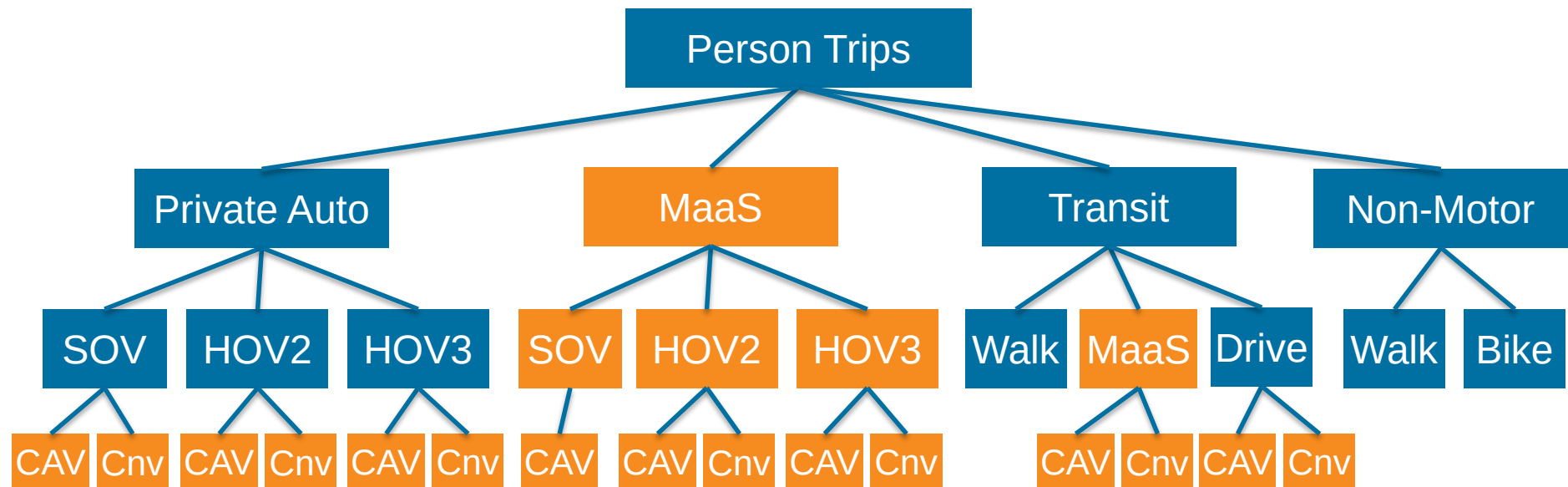
- Add Taxi / TNC mode
- Decrease cost &
- Vary occupancy



Source: Futurism.com

Mode Choice

- Add MaaS mode
- Add CAV / Conventional Submodes
- User specified shares for all modes or only new



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Long-Distance & Trucks



Source: theautofuture.com

- Long distance travelers may use sleeping hours to travel
- There may be more long distance (external) trips as the cost of travel is reduced
- Trucks / long distance travelers may shift to nighttime hours to avoid congestion

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Deadheading / ZOVs

- Types of ZOV trips
 - Private CAVs
 - for car sharing among household members (1)
 - to avoid paid parking
 - by parking at home (2)
 - by parking elsewhere (3)
 - by circulating instead of parking (4)
 - Shared CAVs
 - for passenger pick-up/drop-off (5)
 - to/from depots (6)
(for re-charging / demand response)



Source: driverlesstransportation.com

Private CAV ZOV Modeling Methods

1. for car sharing among household members
 - invert asserted percentage of Os & Ds, gravity model
2. to avoid paid parking by parking at home
 - invert asserted percentage of HB trip ODs for pay TAZs
3. to avoid paid parking by parking elsewhere
 - create trips between pay TAZ & nearest non-pay TAZs as a function of long Ds at pay TAZ
4. to avoid paid parking by circulating instead of parking
 - after assignment, factor up volumes on non-freeway links within buffer of pay TAZ based on short stops



Shared CAV ZOV Modeling Methods

5. for passenger pick-up/drop-off
 - invert all passenger Os & Ds; gravity model
6. to/from depots (for re-charging / demand response)
 - assert depot TAZs with capacities; generate trips by min(difference in demand between periods, charging requirement assumption); gravity between shared CAV Os & Ds and depot TAZ



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Assignment

- Separate autonomous and conventional vehicle classes
- User option to have dedicated CAV-only facilities/lanes and assert high capacities and higher speeds
- User option to assert different capacity consumption in mixed traffic (through PCE factor)





Scenario Planning Example

Burlington, VT Scenario Planning

- Intended to start a conversation about future transportation and traffic implications of the impending CAV future for LRTP
- Not used in selecting projects
- Two scenarios
 - 80% CAV, 50% shared, base occupancy
 - 100% CAV, 65% shared, higher occupancy



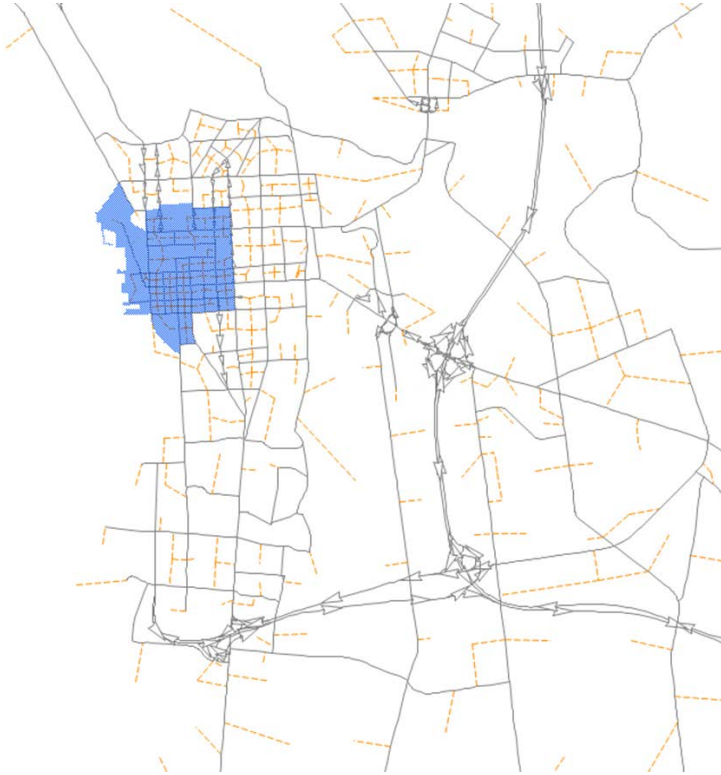
Burlington, VT Scenario Assumptions

- 5% more HBO for induced person trips
- No temporal or destination choice changes
- All ZOV types except intra-household sharing
- Generous increased capacity assumptions
 - Doubling of freeway capacity for 80% scenario
 - Tripling of freeway and 50% increase of arterial capacity for 100% scenario



Burlington, VT

Scenario Assumptions



Pay Parking TAZ



Depot TAZ

Burlington, VT Scenario Results

		CAV (PCE=1.08)	CAV (PCE=1)
	Baseline	80% CAV	100% CAV
Total VMT	5,407,153	7,257,230	5,778,606
Total VHT	156,847	230,545	163,905
VMT/Person	29.5	39.6	31.5
VHT/Person	0.9	1.3	0.9
VMT/Person Trip	5.96	6.02	4.53
VHT/Person Trip	0.24	0.34	0.3
Total Delay (minutes)	1,686,780	3,338,833	1,441,049
Delay/Person	9.21	18.23	7.87
Delay/Vehicle Trip	2.53	3.46	1.79
Average Length of Trip (miles)	8.65	7.52	7.17
Average Length of Trip (minutes)	16.02	14.33	12.20
All Vehicle Trips	666,388	965,533	805,927
ZOV Trips	-	295,009	269,646
% ZOV	-	31%	33%





Conclusions

Parting Thoughts

- Models can be enhanced to capture almost (but not) all dimensions of uncertainty about CAVs
 - Add adjustments to all steps
 - Add taxi/TNC mode
 - Add special ZOV components
- They can NOT tell us what will happen
- They CAN help us understand
 - the range of possible futures
 - the relative importance of different factors
 - the robustness of policies / investments





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